

Shayan Shakeri

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PROFESSIONAL SUMMARY

AI research engineer focused on VLA models, visuomotor manipulation, and sim-to-real deployment. Co-founded Merlyn Labs: 8th place in Stanford's BEHAVIOR-1K Challenge, doubled $\pi 0.5$'s position-swap success with a conservative finetuning recipe (21% to 42%), open-sourced a flow-matching VLA integration for RLinf, and running sim-to-real transfer on a hand-built AlohaMini (merlyn-labs.com/behavior-report). At BMO's AI Centre of Excellence, built deterministic eval pipelines that surfaced systematic bias in a GenAI tool serving \$200B+ AUM.

CORE COMPETENCIES

VLA Models

Imitation Learning & RL

Sim2Real

Robot Manipulation

Policy Evaluation

WORK EXPERIENCE

BMO AI Centre of Excellence Sep 2025 - Present

AI Research Engineer | Toronto, Canada

- Uncovered systematic bias to downplay investment risk in a GenAI tool serving **\$200B+ AUM** wealth management
- Built a **deterministic agent eval pipeline** using hundreds of synthesized inputs to detect misaligned outputs at scale

Merlyn Labs Aug 2025 - Present

AI Researcher & Co-Founder | Toronto, Canada

- Showed $\pi 0.5$ generalization failures on LIBERO-PRO are recipe-induced overfitting; proposed an alternative baseline
- Conservative finetuning doubled position-swap success (**21% to 42%**), showing spatial generalization over memorization
- Open-sourced **flow-matching VLA integration for RLinf**, enabling RL training on the BEHAVIOR-1K suite in OmniGibson
- Running **sim-to-real transfer** of household tasks on hand-built AlohaMini embodiment

Epineuron Technologies May 2021 - Aug 2022

Biomedical & Electrical Engineering Co-op | Mississauga, Canada

- Helped develop **PeriPulse**, an FDA Breakthrough-designated device now in multi-national clinical trials
- Designed and assembled **PCBs** and authored validation protocols (IEC 60601-1, ISO 13485) for clinical test runs
- Optimized power draw via oscilloscope/power analyzer testing, achieving **900%** battery life improvement

RESEARCH & PROJECTS

Stanford BEHAVIOR-1K Challenge 8th Place | Standard Track Sep 2025 - Nov 2025

- Identified **proprioceptive collapse** as critical VLA failure mode; 60% masking improved task success by up to **48%**
- Doubled manipulation success on **long-tail subtasks** by oversampling skill transitions via boundary resampling
- Trained on 10,000+ demonstrations across 22 tasks; published a [technical report](#) and further analysis on [LessWrong](#)

Voice-Controlled Robotic Prosthetic, University of Toronto Sep 2024 - Mar 2025

- Developed LLM-based control pipeline converting natural language to manipulation sequences for **7-DOF** simulated arm
- Integrated **YOLO** object detection with **LiDAR** depth mapping for 3D localization and grasp planning

EDUCATION

M.Eng, MIE - AI & Robotics | University of Toronto Sep 2024 - Apr 2027 (anticipated)

Coursework: Deep Learning, Reinforcement Learning, AI Applications in Robotics, Healthcare Robotics

B.Eng, Engineering Physics (Co-op) | McMaster University Sep 2018 - Jun 2023

Dean's Honour List. Coursework: Computational Multiphysics, Quantum Computing, Numerical Methods, Signals & Systems

SKILLS & INTERESTS

ML & AI: Python, C/C++, PyTorch, JAX, RL, Imitation Learning, LLM Fine-tuning, Alignment Evaluation, Agents

Robotics & Hardware: VLA Models, OmniGibson, MuJoCo, Flow Matching, Sim2Real, ROS2, PCB Design, 3D Printing

Interests: Literature, Dungeons & Dragons, Volleyball, Twilight Imperium, Piano